

Importance of N_2H_x Reaction Pathways to Optimize Okafor Reduced Mechanism for NH_3/H_2 Flame Chemistry

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June 9, 2026

Research Background

Traditional sources of energy rely on combustion of fossil fuels that produce carbon emissions which have adverse effects on the environment:

- Global Warming
- Climate Change

New carbon-free alternative fuels are needed to prevent exacerbating the effects of carbon pollutants

- The Japanese government seeks to reduce total greenhouse gas emissions by 46% before 2030
- By 2050, Japan envisions its cities to be carbon-neutral

Research Background

Ammonia, NH_3 , is gaining interest among researchers for its potential to meet energy needs in a variety of applications.

Ammonia is a suitable fuel for combustion application because:

- Ammonia combustion produces no CO_2
- Ammonia has a high volumetric energy density
- Easier to store than hydrogen
- Infrastructure to produce and transport ammonia already exists
- Used as H_2 carrier

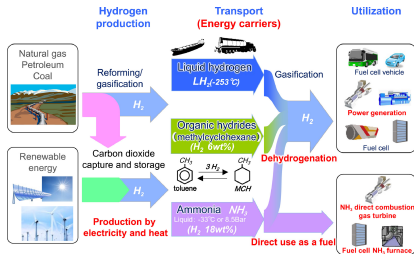


Figure: Hideaki Kobayashi et al. (Jan. 1, 2019). "Science and Technology of Ammonia Combustion". In: *Proceedings of the Combustion Institute* 37.1, pp. 109–133. ISSN: 1540-7489. DOI: 10.1016/j.proci.2018.09.029

Objectives

- Validate Abe Mechanism, at varying combustion conditions, for fuels:
 - $\text{NH}_3/\text{H}_2/\text{Air}$
 - $\text{CH}_4/\text{NH}_3/\text{Air}$
- Flame properties being evaluated:
 - Laminar burning velocity
 - Specie profiles
 - Intermediate species profiles
- Optimize reaction mechanism to fit experimental data for all conditions above satisfactorily

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Prior Findings — $\text{NH}_3/\text{H}_2/\text{Air}$

Abe mechanism reproduces LBV of $\text{NH}_3/\text{H}_2/\text{Air}$ flame to match experimental data

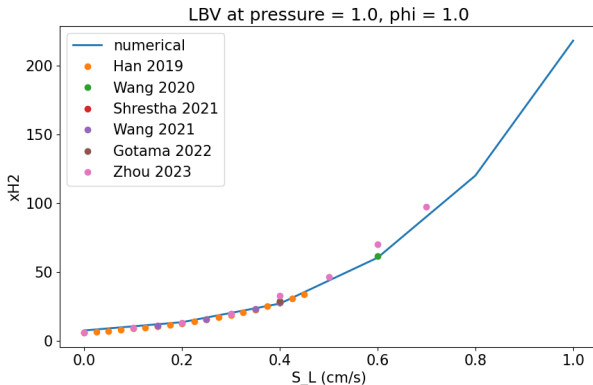


Figure: Laminar burning velocities of $\text{NH}_3/\text{H}_2/\text{Air}$ flames at varying H_2 concentrations while pressure and equivalence ratio is kept constant at 1atm and 1.0 respectively

Prior Findings — CH₄/NH₃/Air

Plotting of experimental data from Okafor 2019¹ was corrected

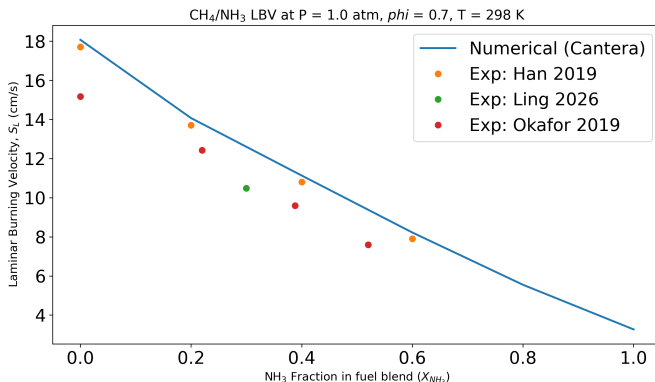


Figure: Laminar burning velocities of CH₄/NH₃/Air flames at varying NH₃ concentrations while pressure and equivalence ratio is kept constant at 1atm and 0.7 respectively

¹Ekenechukwu Chijioka Okafor et al. (June 1, 2019). "Measurement and Modelling of the Laminar Burning Velocity of Methane-Ammonia-Air Flames at High Pressures Using a Reduced Reaction Mechanism". In: *Combustion and Flame* 204, pp. 162–175. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2019.03.008.

Prior Findings — Failed Conditions

- LBV overprediction by Abe mech observed across most CH₄/NH₃/Air conditions.
 - Pressure = 1 atm, $\phi = 1.4$
 - Pressure = 1 atm, $\phi = 1.6$
- Experimental data for higher pressures, 3atm and 5atm, are lacking for both fuels
- Numerical simulations with Okafor reduced mechanism failed to converge
 - Unable to compare LBV from Okafor Reduced Mechanism and Abe mech at the conditions discussed

Next Steps

- Validating Abe mech's specie profiles with experimental data from literature
- Speciation data for relevant fuels and corresponding fuels selected
 - CH₄/NH₃: NH₃, NO, CO
 - NH₃: OH, NH, NO
 - NH₃/H₂: NH₂, NH, NO_x
- For each speciation dataset, simulation methodology employed in the corresponding paper must be used for validating Abe mech

Specie Profiles — CH₄/NH₃

- From Mendiara and Glarborg 2009, the speciation data is used to validate NH₃, NO and CO profiles
- Plug Flow Reactor
 - Diameter 12 mm
 - Length 120 cm
 - Alumina surface
 - Temperature: 973-1773 K

Table: Experimental conditions that need to be simulated with the plug flow reactor. [Teresa Mendiara and Peter Glarborg \(Oct. 1, 2009\)](#). "Ammonia Chemistry in Oxy-Fuel Combustion of Methane". In: *Combustion and Flame* 156.10, pp. 1937–1949. ISSN: 0010-2180. DOI: 10.1016/j.combustflame.2009.07.006

Run	CH ₄ /ppm	O ₂ /ppm	NH ₃ /ppm	N ₂ /%	ϕ	Residence time/s
1	2515	3479	484	99.35	1.55	1296/ T (K)
2	2513	5036	468	99.20	1.07	1296/ T (K)
3	2508	40062	468	95.69	0.13	1293/ T (K)

Specie Profiles — NH₃

- From Brackmann et al. 2016, speciation data of OH, NH and NO profiles are used to validate Abe mech.
- Stagnation Flame Reactor
 - Steel disk to stabilize the flame
 - Flame stabilized at ~6 mm above burner surface so domain should be > 6 mm
 - Temperature profiles from the experiment used as input
 - GRAD parameter was set to 0.02

Table: Experimental conditions that need to be simulated with the stagnation flame reactor. [Christian Brackmann et al. \(Jan. 1, 2016\)](#). "Structure of Premixed Ammonia + Air Flames at Atmospheric Pressure: Laser Diagnostics and Kinetic Modeling". In: *Combustion and Flame* 163, pp. 370–381. ISSN: 0010-2180. DOI: [10.1016/j.combustflame.2015.10.012](https://doi.org/10.1016/j.combustflame.2015.10.012)

Flame	NH ₃ (L _n /min)	Air (L _n /min)	N ₂ -flame (L _n /min)	N ₂ -coflow (L _n /min)
$\phi = 0.9$	2.62	10.3	0	1.2
$\phi = 1.0$	3.38	12.1	0.87	1.2
$\phi = 1.2$	3.9	11.6	0	1.2

Specie Profiles — NH₃/H₂

- From Alturaifi, Mathieu, and Petersen 2023, speciation data of NH₃, NO, NO_x, N₂O and NH₂ profiles used.
- Premixed Laminar Flame-Speed Reactor/Burner-stabilized flat flame
 - Domain/width of ~ 20 mm
 - Experimental flame's temperature profile used as input

Table: Experimental conditions that need to be simulated. [Sulaiman A. Alturaifi, Olivier Mathieu, and Eric L. Petersen \(Jan. 1, 2023\)](#). "A Shock-Tube Study of NH₃ and NH₃/H₂ Oxidation Using Laser Absorption of NH₃ and H₂O". In: *Proceedings of the Combustion Institute* 39.1, pp. 233–241. ISSN: 1540-7489

Flame	X_{NH_3}	X_{H_2}	X_{O_2}	X_{Ar}	V_0 (cm/s)	Φ	p (mbar)
I	0.25	0.05	0.21	0.48	35.1	1	60
II	0.24	0.07	0.21	0.47	42.1	1	50
III	0.22	0.10	0.21	0.46	42.1	1	50
IV	0.21	0.13	0.21	0.45	42.0	1	50
V	0.22	0.10	0.24	0.43	42.2	0.9	50
VI	0.22	0.10	0.20	0.48	42.5	1.1	50
VII	0.25	0.05	0.21	0.48	23.4	1	90
VIII	0.25	0.05	0.21	0.48	17.6	1	120

Thank You